Mini Project Report

Project Title:

Design a combinational circuit to multiply two 2-bit binary numbers. Generate partial products using AND gates. Combine partial products using half adders and logic gates.

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Semester / Class: **3rd Computer Organization**

Academic Year: **2026 – 2027**

Department of Computer Science & Engineering

Budge Budge Institute of Technology (BBIT)

Certificate

This is to certify that the mini project entitled **Design a combinational circuit to multiply two 2-bit binary numbers. Generate partial products using AND gates. Combine partial products using half adders and logic gates** has been successfully completed by

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Acknowledgement

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Design a combinational circuit to multiply two 2-bit binary numbers. Generate partial products using AND gates. Combine partial products using half adders and logic gates.

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- **Abstract**

Digital multiplication is one of the most fundamental arithmetic operations executed within the Arithmetic Logic Unit (ALU) of modern microprocessors, digital signal processors (DSPs), and embedded controllers. Unlike sequential shift-and-add multipliers that require multiple clock cycles, control state machines, and storage registers, a combinational array multiplier computes the product of two binary operands in a single propagation delay using purely combinational logic gates. This mini project presents the architectural design, Boolean formulation, gate-level implementation, and experimental verification of a 2-bit by 2-bit combinational binary multiplier circuit.

The designed circuit accepts two 2-bit unsigned binary inputs—Multiplicand A (A1 A0) and Multiplier B (B1 B0)—and generates a 4-bit binary product C (C3 C2 C1 C0), accurately representing decimal products from 0 × 0 = 0 (0000 in binary) up to 3 × 3 = 9 (1001 in binary). In Stage 1, four 2-input AND gates generate the bit-wise partial products (P0 = A0·B0, P1 = A1·B0, P2 = A0·B1, and P3 = A1·B1). In Stage 2, two Half Adders (HA1 and HA2, each realized using one 2-input XOR gate and one 2-input AND gate) combine the weighted partial products and propagate the internal carry (K1) to produce the final 4-bit result. The circuit was implemented and validated using both a virtual breadboard prototype in Autodesk Tinkercad (utilizing two 74HC08 Quad AND ICs, one 74HC86 Quad XOR IC, and a 7805 5V regulator) and a custom interactive web-based circuit simulator. Testing across all 16 binary input combinations confirmed 100% accuracy against theoretical expectations.

Keywords: Combinational Multiplier, Partial Products, Half Adder, IC 74HC08 (AND Gate), IC 74HC86 (XOR Gate), Arithmetic Logic Unit (ALU), Computer Organization.

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1. Introduction

In Computer Organization and Architecture (COA), the Arithmetic Logic Unit (ALU) serves as the core execution engine responsible for performing arithmetic and bit-wise logical operations on binary words. Among the fundamental arithmetic operations—addition, subtraction, multiplication, and division—hardware multiplication plays a decisive role in determining the overall execution speed of numerically intensive applications such as Digital Signal Processing (DSP), image convolution, matrix transformations, cryptography, and real-time control systems.

At the machine level, binary multiplication of unsigned integers follows the exact same positional shift-and-add principle as paper-and-pencil decimal multiplication. Given a 2-bit Multiplicand A = A1 A0 and a 2-bit Multiplier B = B1 B0, each bit of the multiplier multiplies the entire multiplicand to produce a shifted row of partial products. Because binary variables take only the logic values 0 or 1, the multiplication of any two single bits ($A_i \times B_j$) follows the truth table of a 2-input logical AND gate ($0\times0 = 0$, $0\times1 = 0$, $1\times0 = 0$, and $1\times1 = 1$). Consequently, all partial products in a combinational multiplier can be generated simultaneously in a single gate delay using an array of AND gates.

Once the partial products are generated, terms sharing the same binary positional weight ($2^0 = 1$, $2^1 = 2$, $2^2 = 4$, and $2^3 = 8$) are summed column by column. Since the maximum value of a 2-bit unsigned number is 3 (11 in binary), the maximum possible product of two 2-bit numbers is $3 \times 3 = 9$ (1001 in binary), which requires exactly 4 output bits (C3 C2 C1 C0). In a 2-bit by 2-bit multiplier, this column addition requires only two Half Adders (HA1 and HA2), making it an ideal foundational circuit for understanding parallel hardware arithmetic.

2. Problem Statement

Software-based multiplication loops and sequential shift-and-add hardware multipliers require multiple clock cycles to compute a single product, creating a performance bottleneck in high-speed arithmetic pipelines. The problem addressed in this mini project is to design, implement, and verify a purely combinational digital logic circuit that multiplies two 2-bit unsigned binary numbers (A = A1 A0 and B = B1 B0) and outputs their 4-bit binary product (C = C3 C2 C1 C0) instantaneously without clock synchronization or sequential registers.

Core Design Constraints Specified in the Project Problem:

1. Input Specification: Two 2-bit binary operands — Multiplicand A (A1, A0) and Multiplier B (B1, B0), representing decimal values 0 to 3.
2. Partial Product Generation: Generate all four bit-wise partial products (P0 = A0·B0, P1 = A1·B0, P2 = A0·B1, and P3 = A1·B1) in parallel using four 2-input AND gates.
3. Partial Product Summation: Combine the intermediate partial products and internal carry using two Half Adders (each implemented with a 2-input XOR gate for Sum and a 2-input AND gate for Carry).
4. Output Specification: Produce a 4-bit binary product C3 C2 C1 C0 (decimal range 0 to 9) verified across all 16 truth-table combinations.

3. Objectives

- **Architectural Understanding:** To study the internal dataflow and gate-level organization of combinational array multipliers used in modern processor ALUs.
- **Parallel Partial Product Generation:** To formulate and construct the 4-gate AND array that computes bit-wise partial products P0, P1, P2, and P3 concurrently.
- **Half-Adder Reduction Network:** To design and interconnect two Half Adder stages (HA1 and HA2) using basic XOR and AND gates to sum same-weight partial products and propagate the internal carry bit K1.
- **Virtual Breadboard Hardware Prototyping:** To assemble and test the physical IC layout in Autodesk Tinkercad Circuits using two 74HC08 Quad AND ICs, one 74HC86 Quad XOR IC, a 7805 5V voltage regulator, slide switches, and LED outputs.
- **Interactive Schematic Simulation:** To build a custom interactive web-based circuit simulator capable of visualizing real-time signal propagation, block-level vs. gate-level schematics, and the complete 16-row truth table.

4. Literature Survey / Existing System

In digital computer architecture literature (M. Morris Mano, Computer System Architecture; Carl Hamacher et al., Computer Organization), binary multiplication circuits are broadly classified into two architectural paradigms: Sequential Shift-and-Add Multipliers and Combinational Array Multipliers.

In the existing sequential multiplication approach, the hardware consists of an n-bit adder, an accumulator register, a multiplier shift register, and a sequential control unit. During each clock cycle, the least significant bit of the multiplier register is inspected: if it is 1, the multiplicand is added to the accumulator; the combined register is then shifted right by one bit position. While this sequential approach minimizes gate count for large word lengths, it requires n clock cycles (plus control overhead) to finish a single multiplication.

By contrast, the Combinational Array Multiplier (also known as the Braun Multiplier for unsigned operands) eliminates registers and clock cycles altogether by replicating the addition hardware spatially. Every partial product is generated simultaneously by an AND gate array, and dedicated adders sum each positional column in a feed-forward combinational network. Table 4.1 summarizes the architectural comparison between sequential multipliers and the proposed combinational multiplier.

Architectural Parameter	Sequential Shift-and-Add Multiplier	Proposed 2-Bit Combinational Multiplier
Circuit Classification	Sequential logic circuit (clock-driven)	Purely combinational logic circuit (clockless)
Memory / Registers	Requires Accumulator, Shift Register & Flip-Flops	Zero memory elements or flip-flops required
Execution Latency	Requires N clock cycles + register setup time	Single propagation delay (3 logic gate levels)
Control Unit Complexity	Requires Finite State Machine (FSM) controller	No control unit needed; direct dataflow
Hardware Gate Count (2×2)	1 Adder + 3 Registers + Clock/Control logic	Exactly 6 AND gates + 2 XOR gates (3 DIP ICs)
Primary Application	Area-constrained low-speed controllers	High-speed ALU pipelines & DSP multiply units

5. System Requirements

5.1 Hardware Requirements

The hardware requirements encompass both the electronic components required to construct the 2-bit binary multiplier on a breadboard and the minimum host computer configuration needed to run the simulation tools.

Sl. No.	Component / Equipment	Specification / IC Part No.	Quantity	Role in Circuit
1	Quad 2-Input AND Gate IC	74HC08 / 74LS08 (14-Pin DIP)	2 Nos.	4 gates for P0..P3; 2 gates for HA1 & HA2 Carry
2	Quad 2-Input XOR Gate IC	74HC86 / 74LS86 (14-Pin DIP)	1 No.	2 gates for HA1 Sum (C1) & HA2 Sum (C2)
3	Voltage Regulator & Battery	LM7805 (+5.00 V) + 9V Battery	1 Set	Provides regulated +5.00 V DC VCC supply
4	Input Slide Switches	SPDT Breadboard Slide Switches	4 Nos.	Toggles binary inputs A1, A0, B1, B0 (0V / 5V)
5	Output LEDs & Resistors	Green LEDs + 220 Ω Resistors	4 Sets	Displays 4-bit binary output C3, C2, C1, C0
6	Solderless Breadboard	830 Tie-Points with Power Rails	1 No.	Mounting ICs, switches, and jumper interconnections
7	Host PC (Simulation)	Intel Core i3 / Ryzen 3, 4 GB RAM	1 System	Running Tinkercad & Web Circuit Simulator

5.2 Software Requirements

Sl. No.	Software / Tool Category	Tool / Technology Used	Purpose in Project
1	EDA & Breadboard Simulator	Autodesk Tinkercad Circuits	Virtual breadboard wiring, 5V power regulation & IC testing
2	Interactive Logic Simulator	HTML5, Tailwind CSS, SVG, JavaScript (ES6)	Custom real-time gate-level & block-level circuit simulator
3	Schematic Drafting Engine	Python 3 (Matplotlib, Pillow)	High-resolution IEEE block & gate schematic diagrams
4	Web Browser & OS	Chrome / Edge on Windows 10/11 or Linux	Executing the interactive simulator and Tinkercad workspace

6. System Design

6.1 Mathematical Formulation & Partial Product Matrix

Let the two 2-bit unsigned binary numbers be Multiplicand A = A1 A0 and Multiplier B = B1 B0, where A0 and B0 represent the Least Significant Bits (LSBs, weight 2^0 = 1) and A1 and B1 represent the Most Significant Bits (MSBs, weight 2^1 = 2). Their decimal equivalents are:

> **Binary-to-Decimal Weight Representation & Long Multiplication:**
>
> • Multiplicand: A = (A1 × 2^1) + (A0 × 2^0) = 2·A1 + A0
>
> • Multiplier: B = (B1 × 2^1) + (B0 × 2^0) = 2·B1 + B0
>
> • Product: C = A × B = (2·A1 + A0) × (2·B1 + B0) = 4·(A1·B1) + 2·(A1·B0 + A0·B1) + 1·(A0·B0)
>
> • Output Word: C = (C3 × 2^3) + (C2 × 2^2) + (C1 × 2^1) + (C0 × 2^0) = 8·C3 + 4·C2 + 2·C1 + C0

Arranging the bit-by-bit multiplication in positional columns (2^3, 2^2, 2^1, 2^0) yields the partial product addition matrix shown in Table 6.1. Each partial product Pi is formed by a 2-input AND operation.

Multiplication Stage / Row	Bit Weight 2^3 (8)	Bit Weight 2^2 (4)	Bit Weight 2^1 (2)	Bit Weight 2^0 (1)
Multiplicand (A)			A1	A0
Multiplier (× B)			B1	B0
Partial Product Row 1 (A × B0)			P1 = A1 · B0	P0 = A0 · B0
Partial Product Row 2 (A × B1, shifted left)		P3 = A1 · B1	P2 = A0 · B1	0
Internal Carry from Column 2^1 (HA1 Carry)		K1 = P1 · P2	← (Carry Out)	
Final 4-Bit Binary Product (C3 C2 C1 C0)	C3 = K1 · P3	C2 = K1 ⊕ P3	C1 = P1 ⊕ P2	C0 = P0 = A0 · B0

6.2 Architectural Block Diagram (AND Gates & Half Adders)

As shown in Table 6.1, the hardware architecture naturally divides into two combinational stages:

- **Stage 1 — Partial Product Generator (4 × AND Gates):** Four 2-input AND gates receive inputs from the 4-wire input bus (A1, A0, B1, B0) and compute P0 = A0·B0, P1 = A1·B0, P2 = A0·B1, and P3 = A1·B1 in parallel.

- **Stage 2 — Partial Product Summation (2 × Half Adders):** Because column 2^0 contains only a single term (P0), it passes directly to output C0. Column 2^1 contains two bits (P1 and P2), which are added by Half Adder 1 (HA1) to produce Sum C1 and Carry-out K1. Column 2^2 adds the carry K1 from HA1 to partial product P3 using Half Adder 2 (HA2), producing Sum C2 and Carry-out C3 (weight 2^3).

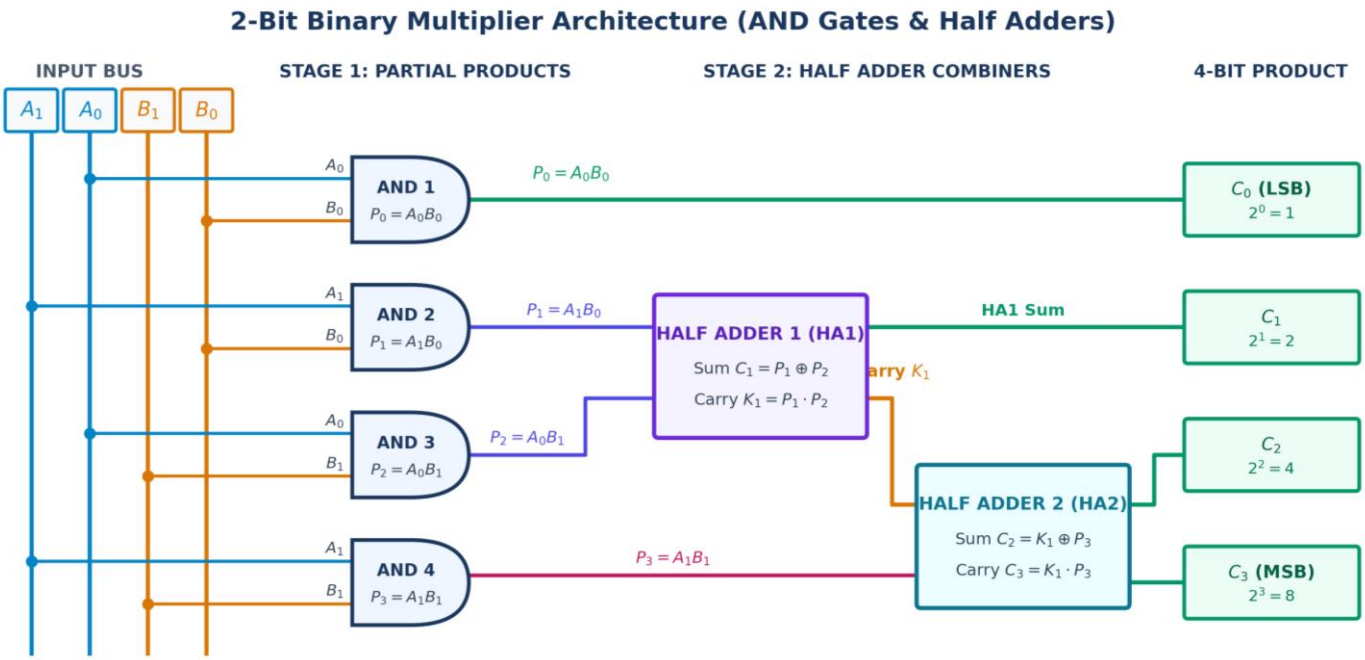


Figure 6.1: Architectural Block Diagram of the 2-Bit Binary Multiplier using 4 AND Gates and 2 Half Adders (HA1, HA2).

6.3 Gate-Level Logic Schematic & Boolean Equations

A Half Adder (HA) is a combinational arithmetic building block that adds two 1-bit binary inputs (X and Y) to produce a Sum bit (S = X ⊕ Y, implemented with a 2-input XOR gate) and a Carry bit (C = X · Y, implemented with a 2-input AND gate). Expanding both Half Adders (HA1 and HA2) into their constituent logic gates yields a total hardware requirement of 6 two-input AND gates and 2 two-input XOR gates.

> **Complete Gate-Level Boolean Equations for Outputs C0, C1, C2, and C3:**
>
> 1. Output C0 (LSB, Weight 2^0): C0 = P0 = A0 · B0
> 2. Output C1 (Weight 2^1): C1 = P1 ⊕ P2 = (A1 · B0) ⊕ (A0 · B1) [HA1 Sum — XOR Gate 1]
> 3. Internal Carry K1 (to 2^2): K1 = P1 · P2 = (A1 · B0) · (A0 · B1) = A1 · A0 · B1 · B0 [HA1 Carry — AND Gate 5]
> 4. Output C2 (Weight 2^2): C2 = K1 ⊕ P3 = ((A1 · B0) · (A0 · B1)) ⊕ (A1 · B1) [HA2 Sum — XOR Gate 2]
> 5. Output C3 (MSB, Weight 2^3): C3 = K1 · P3 = ((A1 · B0) · (A0 · B1)) · (A1 · B1) = A1 · A0 · B1 · B0 [HA2 Carry — AND Gate 6]

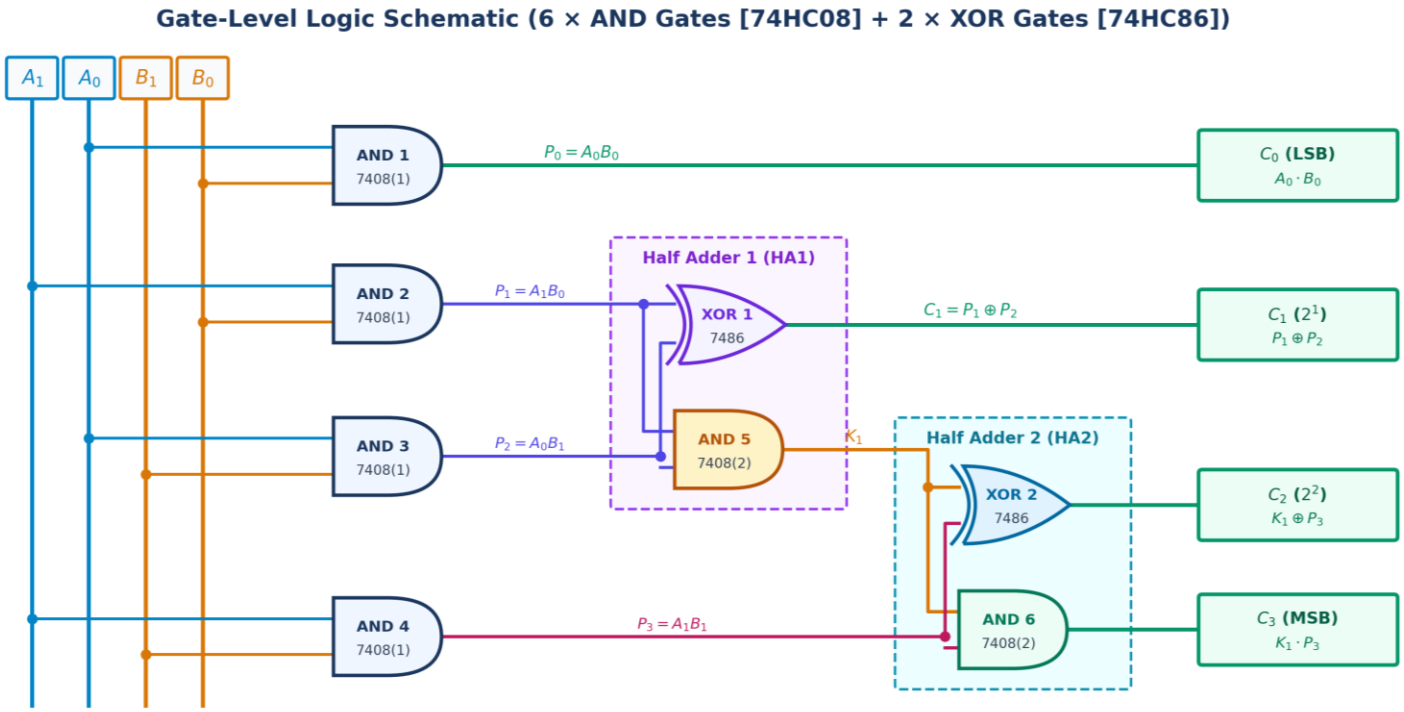


Figure 6.2: Gate-Level Logic Schematic showing all 6 AND Gates (two 74HC08 ICs) and 2 XOR Gates (one 74HC86 IC).

Website URL: bit-binary-multiplier.web.app

GitHub URL: github.com/parthapb45/2-bit-Binary-Multiplier-Combinational

7. Implementation

7.1 Virtual Breadboard Implementation in Autodesk TinkercadTo validate the physical hardware realizability of the 2-bit binary multiplier, the circuit was constructed on a virtual solderless breadboard in Autodesk Tinkercad Circuits. As illustrated in Figure 7.1, the circuit is powered by a 9V DC battery stepped down through an LM7805 5V linear voltage regulator, supplying a rock-solid +5.00 V DC (verified by the attached digital multimeter) to the breadboard's positive power rails.

- **Input Stage (4 Red Slide Switches):** Four SPDT slide switches along the bottom of the breadboard provide the binary inputs A1, A0, B1, and B0. Switching a toggle upward connects the pin to +5.00 V (Logic 1), whereas switching it downward connects to Ground (Logic 0).
- **IC Layout (Left to Right):** The first 14-pin DIP chip on the left is IC1 (74HC08 Quad 2-Input AND Gate), which utilizes all four internal AND gates to generate P0, P1, P2, and P3. The middle chip is IC2 (second 74HC08 Quad 2-Input AND Gate), which uses two of its internal AND gates to compute the carry K1 of Half Adder 1 and the carry C3 of Half Adder 2. The right chip is IC3 (74HC86 Quad 2-Input XOR Gate), which uses two internal XOR gates to compute the sum C1 of Half Adder 1 and the sum C2 of Half Adder 2.
- **Output Stage (4 Green LEDs with Resistors):** Four green LEDs along the top of the breadboard represent the 4-bit product C3, C2, C1, and C0 (ordered from left to right as MSB to LSB), each protected by a 220 Ω series current-limiting resistor to Ground.

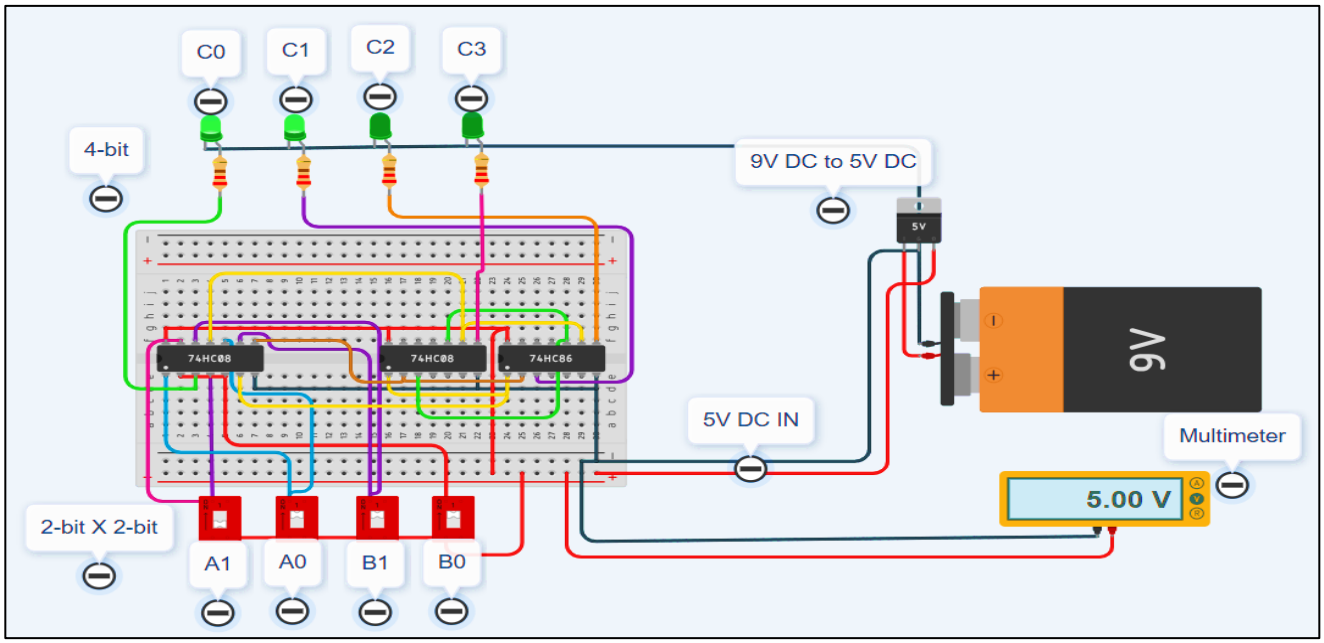


Figure 7.1: Virtual Breadboard Implementation in Autodesk Tinkercad using two 74HC08 ICs, one 74HC86 IC, 7805 5V Regulator (5.00 V), 4 Input Switches, and 4 Output LEDs.

7.2 Custom Interactive Web-Based Circuit Simulator Implementation

In addition to the breadboard prototype, a dedicated interactive web application simulator was developed using HTML5, Tailwind CSS, scalable vector graphics (SVG), and JavaScript. The simulator allows students and evaluators to toggle input bits (A1, A0, B1, B0) or click quick presets (such as 2 × 3 = 6 or 3 × 3 = 9) and immediately observe:

- **Dual Schematic Views:** Users can switch seamlessly between the Half-Adder Block View (4 AND gates + 2 Half Adder blocks) and the Gate-Level View (6 AND gates + 2 XOR gates) with animated glowing green wires indicating Logic 1 signal flow.

- **Live Paper-and-Pencil Matrix & LED Array:** Displays the real-time binary partial product rows (PP1, PP2), internal HA1 carry (K1), and illuminated 4-bit output LEDs (C3, C2, C1, C0).
- –**Interactive Gate Inspector & 16-Row Truth Table:** Clicking any gate displays its live Boolean calculation, while clicking any row in the 16-row truth table loads that input state into the circuit.

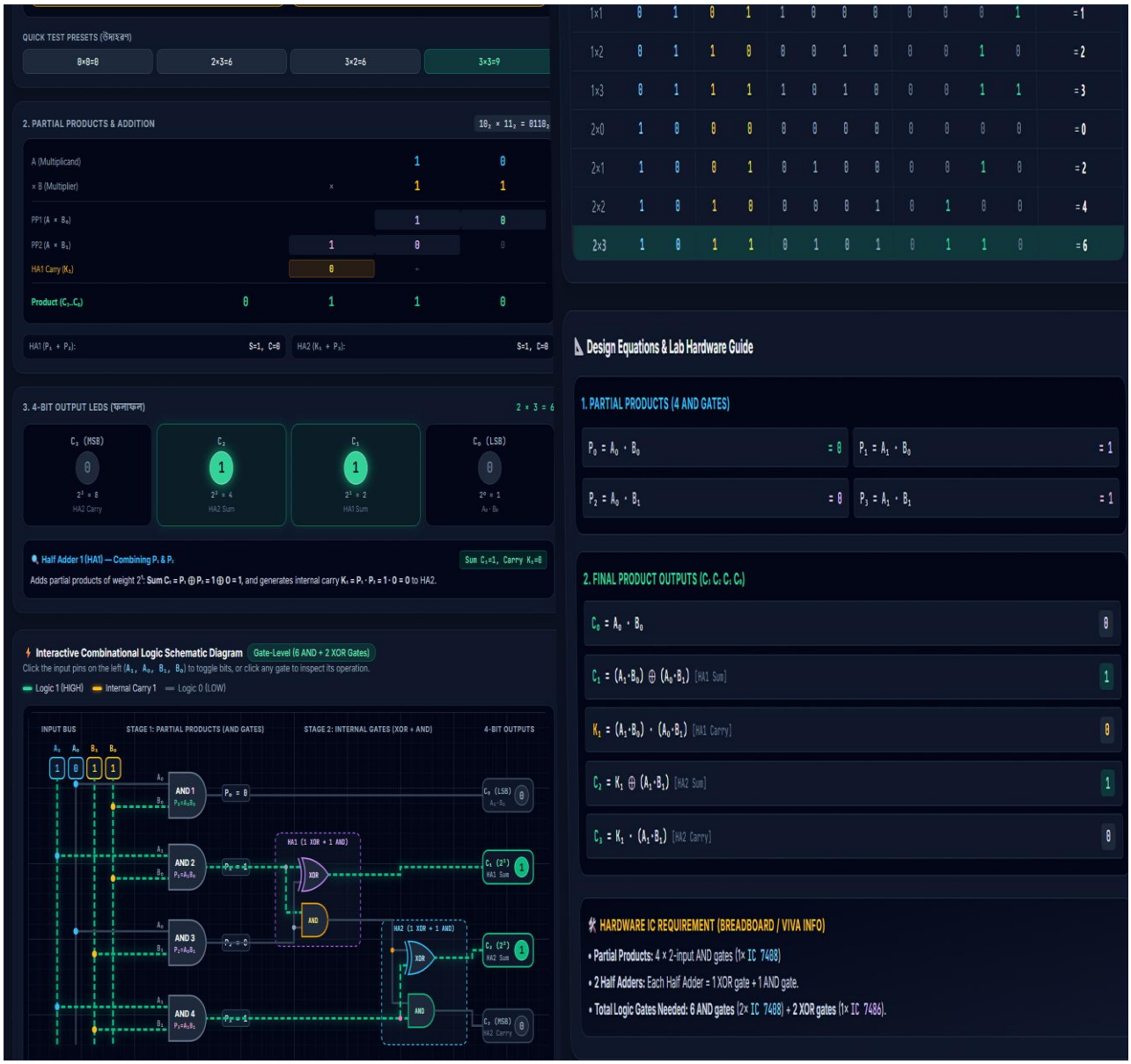


Figure 7.2: Custom Interactive 2-Bit Binary Multiplier Web Simulator demonstrating input A = 2 (10_2), B = 3 (11_2), active Gate-Level schematic, and highlighted Row #11 (Product = 0110_2 = 6).

Website URL: bit-binary-multiplier.web.app

GitHub URL: github.com/parthapb45/2-bit-Binary-Multiplier-Combinational

8. Results & Discussion

8.1 Complete 16-Row Truth Table Verification

Since the circuit has 4 binary input lines (A1, A0, B1, B0), there are 2^4 = 16 distinct input combinations. Table 8.1 presents the complete experimental and theoretical truth table, listing the decimal operands, binary inputs, intermediate partial products (P0, P1, P2, P3), Half Adder 1 internal carry (K1), final 4-bit binary outputs (C3, C2, C1, C0), and decimal product.

Row	Dec	A1	A0	B1	B0	P0	P1	P2	P3	K1	C3	C2	C1	C0	Out
0	0×0	0	0	0	0	0	0	0	0	0	**0**	**0**	**0**	**0**	**0**
1	0×1	0	0	0	1	0	0	0	0	0	**0**	**0**	**0**	**0**	**0**
2	0×2	0	0	1	0	0	0	0	0	0	**0**	**0**	**0**	**0**	**0**
3	0×3	0	0	1	1	0	0	0	0	0	**0**	**0**	**0**	**0**	**0**
4	1×0	0	1	0	0	0	0	0	0	0	**0**	**0**	**0**	**0**	**0**
5	1×1	0	1	0	1	1	0	0	0	0	**0**	**0**	**0**	**1**	**1**
6	1×2	0	1	1	0	0	0	1	0	0	**0**	**0**	**1**	**0**	**2**
7	1×3	0	1	1	1	1	0	1	0	0	**0**	**0**	**1**	**1**	**3**
8	2×0	1	0	0	0	0	0	0	0	0	**0**	**0**	**0**	**0**	**0**
9	2×1	1	0	0	1	0	1	0	0	0	**0**	**0**	**1**	**0**	**2**
10	2×2	1	0	1	0	0	0	0	1	0	**0**	**1**	**0**	**0**	**4**
11	2×3	1	0	1	1	0	1	0	1	0	**0**	**1**	**1**	**0**	**6**
12	3×0	1	1	0	0	0	0	0	0	0	**0**	**0**	**0**	**0**	**0**
13	3×1	1	1	0	1	1	1	0	0	0	**0**	**0**	**1**	**1**	**3**
14	3×2	1	1	1	0	0	0	1	1	0	**0**	**1**	**1**	**0**	**6**
15	3×3	1	1	1	1	1	1	1	1	1	**1**	**0**	**0**	**1**	**9**

8.2 Verification of Key Test Cases & Propagation Delay Analysis

- **Test Case 1 — Multiplying 2 × 3 = 6 (A = 10_2, B = 11_2, Row #11):** As shown in both the Tinkercad breadboard (Figure 7.1) and the Web Simulator (Figure 7.2), when A1=1, A0=0 and B1=1, B0=1, the partial products are P0 = 0·1 = 0, P1 = 1·1 = 1, P2 = 0·1 = 0, and P3 = 1·1 = 1. Half Adder 1 computes C1 = 1 ⊕ 0 = 1 and Carry K1 = 1 · 0 = 0. Half Adder 2 computes C2 = 0 ⊕ 1 = 1 and Carry C3 = 0 · 1 = 0. Thus, the output is C3 C2 C1 C0 = 0110_2 = 6, illuminating the middle two LEDs (C2 and C1).
- **Test Case 2 — Maximum Product 3 × 3 = 9 (A = 11_2, B = 11_2, Row #15):** When all inputs are HIGH (A1=1, A0=1, B1=1, B0=1), all four partial products become 1 (P0=1, P1=1, P2=1, P3=1). Consequently, C0 = 1. In HA1, adding P1=1 and P2=1 yields Sum C1 = 1 ⊕ 1 = 0 and Carry K1 = 1 · 1 = 1. In HA2, adding K1=1 and P3=1 yields Sum C2 = 1 ⊕ 1 = 0 and Carry C3 = 1 · 1 = 1. The final output is C3 C2 C1 C0 = 1001_2 = 9, which is the only state where MSB C3 becomes 1.

- **Critical Path Propagation Delay:** The worst-case combinational delay occurs along the path from Input A1/B0 → Partial Product AND Gate (t_AND) → HA1 Carry AND Gate (t_AND) → HA2 Sum XOR Gate (t_XOR). Using 74HC-series CMOS gates (typical gate delay ~8 ns at 5V), the total multiplication completes in approximately 24 ns (3 gate delays), achieving a theoretical throughput exceeding 40 million multiplications per second.

9. Conclusion & Future Scope

In this mini project, a 2-bit by 2-bit combinational binary multiplier circuit was successfully designed, mathematically derived, implemented on a virtual IC breadboard in Autodesk Tinkercad, and demonstrated through an interactive web-based schematic simulator. By generating all four partial products in parallel with four 2-input AND gates and combining them through two Half Adders (two XOR gates and two AND gates), the circuit achieves instantaneous single-pass binary multiplication across all 16 input states without requiring clock signals or sequential registers.

- **Current Limitations:** The current architecture is designed for 2-bit unsigned binary integers (maximum output 9). As operand bit-width scales to n bits, a standard array multiplier requires n^2 AND gates and n(n-2) Full Adders plus n Half Adders, where carry propagation ripples across adder stages.
- **Future Enhancements:** This 2-bit multiplier can be used as a modular building block to construct a 4×4 or 8×8 recursive multiplier. Furthermore, for high-speed 32-bit and 64-bit processor ALUs, the partial-product reduction stage can be upgraded using Wallace Tree or Dadda Tree carry-save adders, and extended to handle signed 2's complement numbers using the Baugh-Wooley or Modified Booth multiplication algorithm.

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- **[5]** Texas Instruments, SN74HC08 Quadruple 2-Input Positive-AND Gates & SN74HC86 Quadruple 2-Input Exclusive-OR Gates Datasheets, Dallas, TX, USA.

11. Appendix

11.1 IC Pinout & Breadboard Connection Mapping Table

For all three 14-pin DIP Integrated Circuits (IC1: 74HC08, IC2: 74HC08, and IC3: 74HC86), Pin 14 is connected to +5.00 V (VCC) from the LM7805 regulator and Pin 7 is connected to Ground (GND). Table 11.1 provides the exact gate-by-gate pin wiring used in our implementation.

IC Chip	Gate / Stage Role	Input Pin A	Input Pin B	Output Pin	Signal Equation
IC1: 74HC08 #1	AND Gate 1 (Partial Product P0)	Pin 1 (A0)	Pin 2 (B0)	Pin 3 → LED C0	C0 = P0 = A0 · B0
IC1: 74HC08 #1	AND Gate 2 (Partial Product P1)	Pin 4 (A1)	Pin 5 (B0)	Pin 6 (P1)	P1 = A1 · B0
IC1: 74HC08 #1	AND Gate 3 (Partial Product P2)	Pin 9 (A0)	Pin 10 (B1)	Pin 8 (P2)	P2 = A0 · B1
IC1: 74HC08 #1	AND Gate 4 (Partial Product P3)	Pin 12 (A1)	Pin 13 (B1)	Pin 11 (P3)	P3 = A1 · B1
IC3: 74HC86	XOR Gate 1 (Half Adder 1 Sum)	Pin 1 (P1)	Pin 2 (P2)	Pin 3 → LED C1	C1 = P1 ⊕ P2
IC2: 74HC08 #2	AND Gate 5 (Half Adder 1 Carry)	Pin 1 (P1)	Pin 2 (P2)	Pin 3 (Carry K1)	K1 = P1 · P2
IC3: 74HC86	XOR Gate 2 (Half Adder 2 Sum)	Pin 4 (K1)	Pin 5 (P3)	Pin 6 → LED C2	C2 = K1 ⊕ P3
IC2: 74HC08 #2	AND Gate 6 (Half Adder 2 Carry)	Pin 4 (K1)	Pin 5 (P3)	Pin 6 → LED C3	C3 = K1 · P3

11.2 Verilog HDL Gate-Level Structural Code & Simulation Function

Gate-Level Verilog HDL Description (multiplier_2bit.v):

```
module multiplier_2bit (input [1:0] A, B, output [3:0] C);
   wire P1, P2, P3, K1; ( Partial products & HA1 internal carry)
   and G1 (C[0], A[0], B[0]), G2 (P1, A[1], B[0]), G3 (P2, A[0], B[1]), G4 (P3, A[1], B[1]);
   xor HA1_S (C[1], P1, P2),  HA2_S (C[2], K1, P3); (Half Adder Sum outputs (IC 74HC86)
   and HA1_C (K1,   P1, P2),  HA2_C (C[3], K1, P3); (Half Adder Carry outputs (IC 74HC08)
endmodule
```

11.3 Brief User Manual for Testing the Circuit

- **Step 1 (Power Supply Check):** In Tinkercad Circuits, click 'Start Simulation' and verify that the digital voltmeter across the breadboard power rails reads +5.00 V.
- **Step 2 (Set Binary Inputs):** Toggle the four red slide switches (or click the A1, A0, B1, B0 buttons in the Web Simulator) to set any desired 2-bit Multiplicand A and Multiplier B.
- **Step 3 (Read 4-Bit Output):** Observe the four green LEDs (C3, C2, C1, C0 from left to right) and compute the decimal output as 8·C3 + 4·C2 + 2·C1 + C0, comparing against Table 8.1.

Website URL: bit-binary-multiplier.web.app

GitHub URL: github.com/parthapb45/2-bit-Binary-Multiplier-Combinational

Teacher's signature